



Kinetic modeling for enzymatic hydrolysis of pretreated sugarcane straw



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ABSTRACT

A semimechanistic multi-reaction kinetic model was developed for modeling enzymatic hydrolysis of pretreated sugarcane straw (SCS) at high-solids concentration. The model, including: enzyme adsorption, xylose formation, end-product inhibition and substrate reactivity, was adjusted from a reference condition (enzyme concentration of 10 FPU/g-cellulose and solid loading of 15% w/v, dry basis). A simplified model was also proposed to overcome high parameter correlation and improve model fidelity. Predictive capacity of both simplified and original model was verified for different enzyme concentration (5–60 FPU/g-cellulose), solids loading (10 and 20 %) and end-product sugar inhibition (30 and 60 g/L glucose, 10 g/L cellobiose and 10 g/L xylose). The glucose concentration was predicted with reasonable accuracy by the simplified model, even under critical conditions of sugar-inhibition. The model is able to describe batch saccharification of pretreated SCS at high-solids concentrations.

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1. Introduction

Bioethanol is an alternative fuel produced from biomass with potential to reduce greenhouse gas emissions; in fact the conversion of lignocellulosic crops residues is the most attractive option to get for even larger reductions of greenhouse gas emission in transport and industrial sector [1]. However the availability of both agricultural land and cost-effective conversion technology are the main barriers to produce bioethanol in industrial scale [2].

Brazil is the second largest producer of ethanol with 27.54 billion liters (27% of world production) from sugarcane juice [3]. Actually, ethanol yield per hectare of sugarcane is 6000 L/ha and it could be increased by more than 50% without increasing the cultivated area, if sugarcane residues such as bagasse and straw were used [4].

The biochemical route for the conversion of lignocellulosic biomass to ethanol includes four main stages: pretreatment, saccharification or hydrolysis of polysaccharides present in the substrate, fermentation of sugars, and recovery of ethanol by distillation. Sugar production is the critical step for bioethanol production. Enzymatic hydrolysis is preferred over acid hydrolysis for

its high sugar yield (due to the specificity of enzymes), minimal byproducts formation, low energy requirements, mild operating conditions (pH 4–5 and $T=45\text{--}50\text{ }^{\circ}\text{C}$) and low corrosiveness [5–8]. However the high cost of enzymes [9], the low reaction rate [10], and the difficulty of operating at high solids concentration (>12%) [11] are the main barriers for commercialization of lignocellulosic ethanol through biochemical route.

Several factors related to both enzyme characteristics (adsorption, inhibition, synergism, activity) and substrate characteristics (degree of polymerization, crystallinity, accessible surface area, lignin content) affect the hydrolysis of cellulose, therefore including all of them into a kinetic model to describe the synergistic action of enzymes on lignocellulosic substrates is one of the most challenging subjects in engineering for this process. Many kinetic models have been proposed in the literature to describe the enzymatic hydrolysis of cellulose [12,13]. Such models are mainly motivated by both, demonstrating an understanding of physical process in a mathematical structure that adequately fit the experimental data, and development of model-based applications for process design, simulation and control.

Many efforts have been made to reduce the ethanol cost production, including: improving pretreatment technologies; genetic modifying of plant cell walls, genetically improving the enzyme activity and productivity; applying different production schemes such as separate hydrolysis and fermentation (SHF) and simulta-

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Nomenclature

C	cellulose concentration (g/L)
E_B	concentration of bound enzyme onto solids (g/L)
E_{BC}	concentration of bound enzyme onto cellulose (g/L)
E_{BH}	concentration of bound enzyme onto hemicellulose (g/L)
E_F	free enzyme concentration (g/L)
E_{max}	maximum mass of enzyme that can be absorbed onto a unit mass of substrate (g protein/kg substrate)
E_T	total enzyme concentration (g/L)
G	glucose concentration (g/L)
G_2	cellobiose concentration (g/L)
H	hemicellulose concentration (g/L)
K_{3M}	substrate (cellobiose) saturation constant (g/L)
K_{ad}	dissociation constant for the enzyme adsorption/desorption reaction (L/g protein)
K_{iIG}	inhibition constants for glucose (g/L); $i = 1, 2, 3$, and 4
K_{iIG2}	inhibition constants for cellobiose (g/L); $i = 1, 2$ and 4
K_{iIX}	inhibition constants for xylose (g/L); $i = 1, 2, 3$ and 4
k_{ir}	reaction rate constants (L/g·h); $i = 1, 2$ and 4. k_{3r} (h ⁻¹)
L	lignin concentration (g/L)
r_i	reaction rate (g/L·h); $i = 1, 2, 3$ and 4
R_s	substrate reactivity parameter (dimensionless)
S	solids concentration (g/L)
S_0	initial solid concentration (g/L)
X	xylose concentration (g/L)
α	constant relating substrate reactivity with hydrolysis degree (dimensionless)

neous saccharification and fermentation (SSF); and optimization of the hydrolysis operating conditions. High solids saccharification (>12%) presents potential economic advantage over the conventional hydrolysis at low solids loading such as, lower capital cost due to reduced volume reactor; lower operating cost that result from lower energetic requirements for heating and cooling; lower downstream processing cost because of the higher product concentration and reduced disposal and treatment cost due to lower water consumption [14].

Although the effects of high solids concentration over both enzyme inhibition [15] and slurry viscosity [16] have been studied, few models have been used to describe the enzymatic hydrolysis performance under these special operating conditions. The goal of this work is the development of a semimechanistic kinetic model to describe the complex enzymatic hydrolysis of sugarcane straw at high solids concentration with potential to be used in model-based applications such as simulation and optimization of the production process of lignocellulosic ethanol.

2. Enzymatic hydrolysis models

Cellulose saccharification is a complex heterogeneous reaction catalyzed by the synergistic action of enzyme pool, including three major types of enzymatic activities: endoglucanases, exoglucanases, and β -glucosidases [17]. Many models have been proposed to describe the cellulose enzymatic hydrolysis [12,13] based on the reactional scheme of Fig. 1. Zhang and Lynd [13] classified the enzymatic hydrolysis models based on the number of solubilizing activities and substrate state variables, as: (1) nonmechanistic model; (2) semimechanistic model; (3) functionally based model; and (4) structurally based model. Models based on a defensi-

ble adsorption model and using only the concentration variable to describe the substrate state and/or using a single cellulose activity are called semimechanistic. Many hydrolysis models used for the design of industrial systems are included into this category [21]. Although semimechanistic models can be limited to provide a detailed description about the effects of substrate features (crystallinity, degree of polymerization, surface area) and multiple hydrolyzing activities on enzymatic hydrolysis [13], these models are useful once minimal structural information is required for descriptive purpose such as reactor design and model-based applications such as simulation and process optimization.

An early kinetic model for enzymatic hydrolysis of lignocellulosic substrate (rice straw) was proposed by Wald et al. [5]; enzyme adsorption, product inhibition, and a multi-enzyme system were considered into the model. The substrate was modeled assuming two regions (crystalline and amorphous) with different properties related to the enzyme adsorption and hydrolysis. The enzyme adsorption was expressed as function of available sorption sites and the accessible surface area following the Langmuir isotherm. This model was able to describe enzymatic hydrolysis of rice straw at high substrate (up to 333 g/L) and enzyme concentration (up to 9.2 FPU/mL). The modeling of enzyme adsorption on substrate was neglected by researcher in the beginning and generally a Michaelis–Menten (M–M) model was used to describe the enzymatic hydrolysis, though the substrate is not in excess in this heterogeneous system. Gusakov et al. [18,19] model followed this approach, considering also cellulase composition, structural complexity of cellulose, products inhibition, and enzyme inactivation due to unproductive adsorption to substrate. The model satisfactorily predicted batch glucose and cellobiose concentration in the substrate concentration range of 5–100 g/L and enzyme concentration range of 5–60 g/L.

Philippidis et al. [20] proposed a SSF-based kinetic model to simulate enzymatic hydrolysis of pure cellulose. The kinetic parameters were estimated by nonlinear regression from experimental data. Since neither an adsorption model, enzyme inactivation, nor substrate complexity were considered in this model, the cellulose and cellobiose concentration predictions were overestimated and underestimated, respectively, by the model after 24 h of hydrolysis.

One of the most complete semimechanistic kinetic models for enzymatic hydrolysis of cellulose was proposed by Kadam et al. [9], who developed a multireaction kinetic model for batch enzymatic hydrolysis of corn stover pretreated with dilute acid. This model was the first to include xylose inhibition beside of glucose and cellobiose inhibition and included the direct conversion of cellulose to glucose. The temperature effect and the reactivity substrate (R_s) were also incorporated. Model predictions about the effect of inhibition sugar were successfully described for initial concentrations of glucose (30 and 50 g/kg), cellobiose (30 g/kg) and xylose (30 and 40 g/kg). This model has proved to be useful in the simulation and optimization of enzymatic hydrolysis of lignocellulosic substrate [9,21], and could describe successfully the enzymatic hydrolysis of bagasse and straw of sugarcane.

2.1. Model description

The Kadam et al. model [9] is considered sophisticated enough to describe the enzymatic hydrolysis of lignocellulosics and has been chosen to model the enzymatic hydrolysis of sugarcane straw. This model has been experimentally verified [11,21], evaluated in applications of simulation and optimization process [11,22], and recently used to model the enzymatic hydrolysis of sugarcane bagasse [23]. The proposed model is based on the biochemical of enzymatic hydrolysis and include enzyme adsorption, product inhibition, substrate reactivity, and conversion of hemicellulose to

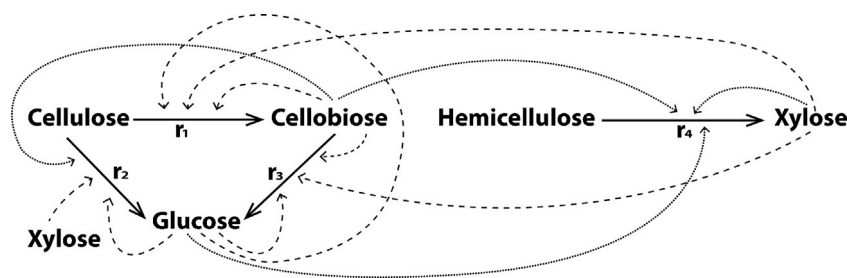


Fig. 1. Scheme for cellulose hydrolysis reaction. Enzymes involved in r_1 : endo- β -1,4-glucanase (EC 3.2.1.4) and exo- β -1,4-cellobiohydrolase (CBH) (EC 3.2.1.91). Enzymes involved in r_2 : exo- β -1,4-cellobiohydrolase (CBH) (EC 3.2.1.91) and exo- β -1,4-glucan glycohydrolase (EC 3.2.1.74). Enzymes involved in r_3 : β -glucosidase. Enzymes involved in r_4 : exo- β -1,4-xylanase. Dashed and dotted lines show the sugars inhibition on enzymes. Both r_4 and dotted line were adopted from [23]. Modified from [4].

xylose according to Câmara [23], nevertheless neglects the thermal and mechanical enzymatic inactivation.

2.1.1. Enzyme adsorption

The enzyme adsorption on the substrate surface to form an enzyme-substrate complex that releases a product is the key step in enzymatic catalysis. Many of early kinetic models of enzymatic hydrolysis were based on the classical kinetic of M–M, however it is only valid when the substrate is in excess relative to the enzyme [17]. Although Langmuir-type isotherms have been used to describe the enzyme adsorption [5,9,21], the Langmuir model assumptions as uniform binding sites and no interaction between the absorbing molecules, are not valid for cellulase adsorption on cellulose; however, the hyperbolic form of Langmuir-type isotherm is useful for mathematical representation of the enzyme adsorption data [9], thus it has been chosen in this work to model the adsorption behavior of the enzyme.

2.1.2. End-product inhibition

Cellulases inhibition by its hydrolysis product (glucose and cellobiose) has long been known. Several works have been conducted to understand the pattern of this inhibition, with different approaches and inconclusive results [13,24]. Competitive mode inhibition assumes that inhibitor sugar is substrate analogue and bind competitively to the active site, retarding the formation of enzyme-substrate complex. It has been suggested that competitive inhibition is dominant and more realistic [9,25,26], therefore, the competitive inhibition pattern have been adopted to represent all inhibitions product (glucose, cellobiose, and xylose) on the enzyme.

2.1.3. Substrate reactivity

Substrate reactivity (R_s) represents the variations on substrate structural features (i.e., crystallinity, degree polymerization, surface area, substrate accessibility) during hydrolysis. Cellulose is formed by both crystalline and amorphous regions and some researchers have found that amorphous cellulose is degraded 3–30 times faster with respect to highly crystalline cellulose by both fungal cellulase and ruminal bacteria [17,27,28], this suggests that R_s is an important factor affecting the enzymatic hydrolysis of cellulose.

However, as indicated by Zheng [29], other researchers did not find substantial changes with hydrolysis progress or obtained equivocal results. Since it is not easy to obtain independent measures of enzyme adsorption equilibria for crystalline and amorphous cellulose, in fact it is not necessary to make this distinction to develop an effective kinetic model. Some previous studies have successfully used R_s to model the structural change of the lignocellulosic substrate [9,21]. Thus, all substrate transformations were lumped into one parameter R_s correlated with the normalized substrate concentration (S/S_0) according to Eq. (4) in Table 2, where the dimensionless constant of proportionality (α) was experimentally determined as close to unit for other lignocellulosics substrates [9,29], and was also assumed as unit in this work based on previous modeling of enzymatic hydrolysis of sugarcane bagasse [23].

3. Materials and methods

3.1. Pretreatment and enzymes

The SCS kindly provided by the Itapira Mill (Catanduva, São Paulo, Brazil), was initially milled using a laboratory knife mill (type-Wiley) (SP-30, SPlabor, Presidente Prudente, SP, Brazil) to a particle size smaller than 4 mm. Liquid hot water pretreatment was performed in a laboratory-scale stirred reactor (model 4532, Parr, Co., Moline, IL) with 1/10 (w/v) solid/liquid ratio of dry SCS loading at 195 °C for 10 min. The solid fraction was washed with water to remove the solubilized contents, and then was used as a substrate in the enzymatic hydrolysis tests. The pretreated SCS was composed of 66% (w/w, dry basis) cellulose, 8.3% (w/w, dry basis) hemicellulose and 25.2% insoluble lignin (w/w, dry basis) contents. The commercial enzymatic complex Cellic CTec2 (Novozymes, Araucária, Paraná, Brazil) with activity of 203 FPU/mL (36 mg protein/mL) was used in the hydrolysis essays.

3.2. Enzymatic hydrolysis

Pretreated SCS was enzymatically hydrolyzed at 13 different conditions of solids, enzyme and initial sugar loading, as shown

Table 1
Experimental conditions for enzymatic hydrolysis of pretreated SCS.^a

		Experiments					
		Reference condition	Verification conditions				
Experimental conditions			Enzyme loading	Solid loading	Initial glucose	Initial cellobiose	Initialxylose
Enzyme loading(FPU/g-celulose)	10		5,15,20,25,30,60	10	10	10	10
Solid loading (% w/v)	15		15	10,20	15	15	15
Initial glucose (g/L)	0		0	0	30,60	0	0
Initial cellobiose (g/L)	0		0	0	0	10	0
Initial xylose (g/L)	0		0	0	0	0	10

^a Temperature, pH, shaking speed, and hydrolysis time were kept constants at 50 °C, 4.8, 250 rpm and 72 h, respectively.

Table 2

Kinetic equations of the enzymatic hydrolysis model.

Langmuir-type isotherm	$E_{BS} = \frac{E_B}{S} = \frac{E_{max}K_{ad}E_F}{1+K_{ad}E_F}$	Eq. (1)
Enzyme adsorbed on cellulose	$E_{BC} = E_B \frac{C}{S}$	Eq. (2)
Enzyme adsorbed on hemicellulose	$E_{BH} = E_B \frac{H}{S}$	Eq. (3)
Substrate reactivity	$R_S = \alpha \frac{S}{S_0}$	Eq. (4)
Cellulose (C) to cellobiose (G2)	$r_1 = \frac{k_{1r}E_{BC}R_S S}{1 + \frac{G2}{K_{IG2}} + \frac{G}{K_{IG}} + \frac{X}{K_{IX}}}$	Eq. (5)
Cellulose (C) to glucose (G)	$r_2 = \frac{k_{2r}E_{BC}R_S S}{1 + \frac{G2}{K_{IG2}} + \frac{G}{K_{IG}} + \frac{X}{K_{IX}}}$	Eq. (6)
Cellobiose (G2) to glucose (G)	$r_3 = \frac{k_{3r}E_{BH}R_S S}{K_{3M} \left(1 + \frac{G}{K_{3IG}} + \frac{X}{K_{3IX}} \right) + G2}$	Eq. (7)
Hemicellulose (H) to xylose (X)	$r_4 = \frac{k_{4r}E_{BH}R_S S}{1 + \frac{G2}{K_{4IG2}} + \frac{G}{K_{4IG}} + \frac{X}{K_{4IX}}}$	Eq. (8)
Total enzyme	$E_T = E_F + E_B$	Eq. (9)
Total solids	$S = C + H + L$	Eq. (10)
Cellulose balance	$\frac{dC}{dt} = -r_1 - r_2$	Eq. (11)
Cellobiose balance	$\frac{dG2}{dt} = 1.056r_1 - r_3$	Eq. (12)
Glucose balance	$\frac{dG}{dt} = 1.111r_2 + 1.053r_3$	Eq. (13)
Hemicellulose balance	$\frac{dH}{dt} = -r_4$	Eq. (14)
Xylose balance	$\frac{dX}{dt} = 1.136r_4$	Eq. (15)

in Table 1. Experimental data for kinetic parameter estimation was generated from one reference condition, that is, 15% (w/v) of solids, charged to 250 mL flasks; enzyme concentration of 10 FPU/g-cellulose without β -glucosidase supplementation; incubation temperature = 50 °C, shaking speed = 250 rpm, initial pH 5.0, and incubation time = 96 h. Other set of experiments were conducted under conditions out of the reference condition range to verify the model predictions at different enzyme concentrations (5–15 FPU/g-cellulose), solids loading (10 and 20% w/v), and initial sugar concentration: glucose (30 and 60 g/L), xylose (10 g/L) and cellobiose (10 g/L). For this experiments, liquid supernatant samples were collected at sampling time = 0, 1, 2, 6, 12, 24, 48 and 72 h, and analyzed for glucose, xylose and cellobiose concentration using HPLC (Shimadzu, Kyoto, Japan) with a Biorad Aminex HPX-87H column and H₂SO₄, 5 mM as mobile phase. The column temperature was 65 °C and the flow rate was 0.6 mL/min.

The adsorption parameters of the Langmuir-type isotherm were estimated from experimental data for different enzyme loading (5–60 FPU/g-cellulose). The free enzyme was assumed to be the protein concentration in the solution and was measured by the Bradford assay with BSA as a standard (Sigma–Aldrich). The bound enzyme was calculated from enzyme mass balance with the known initial protein concentration.

3.3. Model development

A modified Kadam et al. model, described above in Section 2.1, has been chosen to describe the enzymatic hydrolysis of pretreated SCS. As shown in Fig. 1, a simplified scheme of four reactions is the base for the cellulose hydrolysis modeling. Each reaction is potentially inhibited following the competitive mode, by the product sugars (glucose, cellobiose and xylose). Inhibition by xylose was included in the model based on compounds detected in the hydrolyzed enzymatic. During hydrolysis, enzymes are first adsorbed on the components of the substrate surface (cellulose, hemicellulose and lignin); this adsorption was expressed by a Langmuir-type isotherm. The main assumptions for derivation of the model equations include: (1) enzymatic complex was modeled as a pseudo-enzyme with several enzymatic activities (i.e., a single variable was used to describe the enzyme activity); (2) total enzyme activity was considered proportional to protein concentration in the aqueous phase [30], therefore, free enzyme was measured as the protein concentration in the supernatant; (3) Langmuir-type isotherm describes enzyme adsorption in the reactions rates (r_1 , r_2 and r_4) occurring on the substrate surface; (4) there is no distinction between crystalline and amorphous cellulose, thus cellulose was

considered uniformly susceptible to enzymatic attack; (5) composition of the surface substrate for enzyme adsorption is assumed to be roughly equal to the entire fiber composition, then the composition of the entire fiber was used to calculate the adsorbed enzyme on the surface components [31]; (6) mechanical and thermal enzyme inactivation is not significant; (7) cellobiose conversion to glucose occurs by homogeneous catalysis following the classical M–M kinetics.

According to these assumptions, the model equations in Table 2 were derived. The amount of enzyme adsorbed on substrate and its components is calculated with Eqs. (1)–(3). Substrate reactivity is described by Eq. (4) with α assumed to be equal to 1 based on previous modeling work of hydrolysis of sugarcane bagasse [23]. The rates for the reactions shown in Fig. 1 were calculated using Eqs. (5)–(8). The mass balance for enzyme, substrate components and sugar products are expressed by Eqs. (9)–(15).

3.4. Model simplification

The original model was simplified from the assumption of modeling the enzymatic activities of cellulase and hemicellulase as a pseudo-enzyme able to produce glucose, cellobiose and xylose from the substrate. Based on fundamentals of enzymatic kinetics it was proposed that a single set of inhibition parameters (i.e., K_{IG} , K_{IG2} and K_{IX}) could adequately describe the overall effects by product inhibition for reactions occurring at the solid–liquid interface (i.e., r_1 , r_2 and r_4). Therefore, Eqs. (5)–(6) and (8) of original model (see Table 2) were replaced by Eqs. (16)–(18), respectively, in the simplified model.

$$r_1 = \frac{k_{1r}E_{BC}R_S S}{1 + \frac{G2}{K_{IG2}} + \frac{G}{K_{IG}} + \frac{X}{K_{IX}}} \quad (16)$$

$$r_2 = \frac{k_{2r}E_{BC}R_S S}{1 + \frac{G2}{K_{IG2}} + \frac{G}{K_{IG}} + \frac{X}{K_{IX}}} \quad (17)$$

$$r_4 = \frac{k_{4r}E_{BH}R_S S}{1 + \frac{G2}{K_{IG2}} + \frac{G}{K_{IG}} + \frac{X}{K_{IX}}} \quad (18)$$

Model simplification reduced from 16 to 10 the kinetic parameters of the original model. This subset of parameters were estimated using both reference condition and all available experimental data to assess improvements in model predictions when the model is supplied with more experimental information.

3.5. Computational methodology

The EMSO software (Environment for Modeling, Simulations and Optimization) [32] was used for solving the kinetic model. Weighted Least Squares (WLS) function was minimized to estimate the parameters models using the Flexible Polyhedron Method [33].

3.6. Model parameters estimation

The adsorption parameters for Langmuir-type isotherm were estimated independently from experimental measurements of protein in the supernatant for the hydrolysis conducted under different enzymatic loading (5–60 FPU/g-cellulose), and then these parameters were used as inputs to estimate the kinetic parameters model. The kinetic parameters were estimated for original and simplified model from a reference enzymatic hydrolysis condition [solids loading of 15% (w/v, dry basis) and enzyme concentration of 10 FPU/g-cellulose] and using all experimental data available. Other detailed experimental conditions are listed in Table 1.

Table 3
Estimated kinetic parameters.

Kinetic parameters	Units	In this study				[23] ^c	[9] ^c	[35] ^c	[36] ^c
		Original Model ^a	Original Model ^b	Simplified Model ^a	Simplified Model ^b				
k_{1r}	L/g/h	0.177	0.509	0.080	0.204	0.65	22.3	94.72	1.224
k_{2r}	L/g/h	8.81	12.75	2.02	7.75	3.17	7.18	958.3	19.08
k_{3r}	h ⁻¹	201.0	165.7	1.74	145.7	254.5	285.5	432.16	252
k_{4r}	L/g/h	16.34	13.46	15.8	99.9	57.79		167.27	9.72
K_{1IG2}	g/L	0.402	0.016	0.402	3.37	9.83	0.015	1.00×10^{-5}	0.0014
K_{1IG}	g/L	2.71	0.710	5.12	0.953	0.176	0.1	7.33	0.073
K_{1IX}	g/L	2.15	0.559	39.4	0.936	9.54	0.1	8.92	0.1007
K_{2IG2}	g/L	119.6	89.18				132	7.33	132
K_{2IG}	g/L	4.69	0.551			0.15	0.04	1.15×10^{-3}	0.34
K_{2IX}	g/L	0.095	0.581			8.79	0.2	6.13	0.029
K_{3M}	g/L	26.6	47.20	0.507	118.8	24.5	24.3	0.022	24.3
K_{3IG}	g/L	11.06	0.011	13.3	0.081	41.75	3.9	1.45×10^{-5}	3.9
K_{3IX}	g/L	1.023	110.0	118.2	70.4		201.0	39.19	201
K_{4IG2}	g/L	16.25	30.16						24.3
K_{4IG}	g/L	4.0	8.61			0.023		0.0747	2.39
K_{4IX}	g/L	154.0	134.1			9.81		23.12	201
Relative inhibition effects									
$1_{IG2}/K_{1IG}$		0.15		0.08	3.54	55.85	0.15	1.36×10^{-6}	0.02
K_{2IG2}/K_{2IG}		25.5					3300	6374	388.2
K_{3M}/K_{3IG}		2.41		0.04	1459.7	0.59	6.23	1517	6.23
K_{4IG2}/K_{4IG}		4.06							10.17
K_{1IX}/K_{1IG}		0.79		7.69	0.98	54.20	1	1.22	1.38
K_{2IX}/K_{2IG}		0.02				58.60	5	5330.4	0.09
K_{3IX}/K_{3IG}		0.09		8.89	864.9		51.54	2.70×10^6	51.54
K_{4IX}/K_{4IG}		38.5				426.5		309.5	84.10

^a OM: Original model (with all 16 parameters); OM-AE: Original model adjusted with all experimental data; SM: Simplified model (with 10 parameters); SM-AE: Simplified model adjusted with all experimental data.

^b Kinetic parameter estimated using all available experimental data.

^c k_{ir} ($i = 1, 2, 4$) in (kg/g/h); K_{1IG2} , K_{1IG} , K_{1IX} , ($i = 1, \dots, 4$) and K_{3M} in (g/kg).

3.7. Model verification

Experiments for verification of model predictions were performed for conditions outside the design space with different dry solid loadings (10 and 20% w/v), enzyme concentration (5–60 FPU/g-cellulose), and initial sugar concentration (cellobiose and xylose of 10 g/L, and glucose of 20 and 40 g/L) to assess the ability of the model to describe product inhibition.

4. Results and discussion

The approach followed in this study included: (1) obtaining adsorption equilibrium data from experiments with different enzyme concentration to estimate Langmuir parameters; (2) obtaining experimental data from enzymatic hydrolysis under a reference condition to estimate model kinetic parameters; (3) performing model simulations with the estimated parameters to verify model predictions beyond the set of experimental conditions used for parameter estimation.

4.1. Parameter estimation

Adsorption parameters were estimated via least squares function from enzyme adsorption experiments using non-linear form of a Langmuir-type isotherm (Eq. (1)) to avoid adjustment distortion caused by fitting linear transformations. Adsorption parameters estimated are as follows [34]: $E_{\max} = 8.32 \pm 2.3$ g/kg and $K_{ad} = 7.16 \pm 4.4$ L/g.

Adsorption parameters were used as input to the model for estimating kinetic parameters. Experimental data from measured variables (i.e., glucose, cellobiose and xylose concentration) were used for parameter estimation. Parameters were estimated via WLS function using EMSO software [32]; results are listed in columns 3–6 of Table 3. Simulation results of enzymatic hydrolysis of pre-

treated SCS obtained from fitted model for reference condition are shown in Fig 2A.

Recent works have estimated the kinetic parameters of Kadam et al. [9] model from experimental data of enzymatic hydrolysis for different pretreated biomass, including: corn stover [9], sugarcane bagasse [23], corncob stock [35], and wheat straw [36]. Kadam et al. [9] were the first incorporating xylose inhibition from pretreatment; Cămară [23] considered xylose formation from hemicellulose during saccharification; Flórez-Sánchez et al. [35] extended the model to xylo-oligosaccharides and arabinose formation and inhibition; and Prunescu and Sin [36] included acid acetic formation and furfural inhibition.

A comparison was attempted between kinetic parameters estimated in this study and those from previous publications (columns 7–10 of Table 3). According to the parameters values for the original model, cellobiose was the strongest inhibitor of cellobiose formation (r_1) to all compared studies except for Cămară [23]. Researchers agree that direct glucose formation from cellulose (r_2) was slightly inhibited by cellobiose, most of them found glucose to be the main reaction inhibitor [9,35], however our estimated values for original model suggest that xylose is also a strong inhibitor of cellulase in r_2 , as was found by Prunescu and Sin [36]. Even considering xylose inhibition effect and agreeing about the hierarchical importance of sugar inhibitions in r_2 , the relative inhibition values in Table 3 show important differences of magnitude order between studies (e.g., K_{2IX}/K_{2IG} is more than 1000 times higher for Flórez-Sánchez et al. [35] than Kadam et al. [9]). This seemingly contradictory findings show the difficult to make direct comparison between parameters from different reaction schemes and model structures, since the complex mechanisms of cellulose hydrolysis by cellulase remains unclear. Such comparisons become more inconvenient because different experimental (i.e., enzyme mixtures, pretreated substrates, enzyme and substrate loading) and operational conditions (temperature, pH and reactor system) are used by the researchers [9]. An additional important reason to avoid

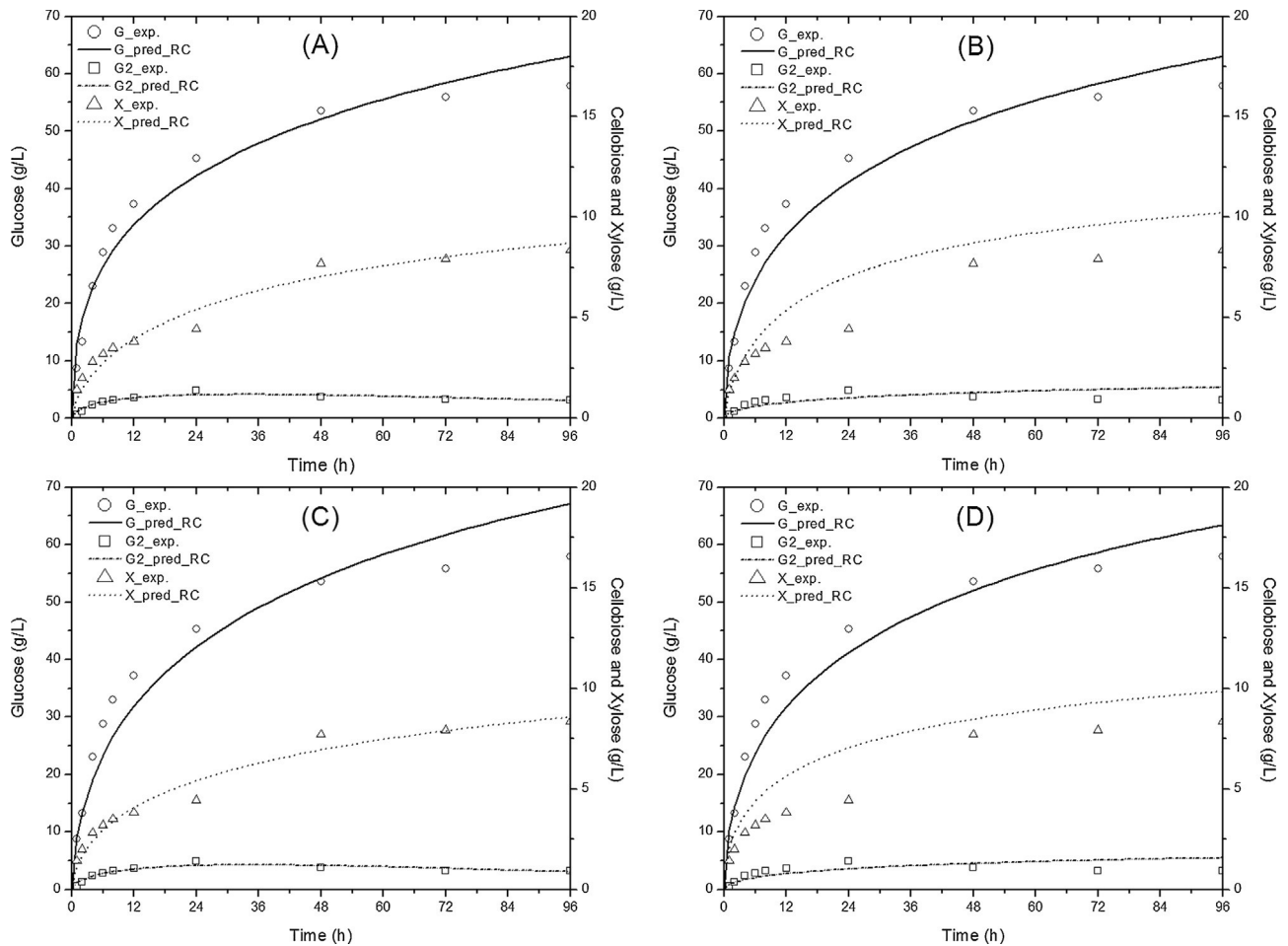


Fig. 2. Enzymatic hydrolysis of pretreated SCS under reference condition. (A) Original model adjusted from reference condition; (B) original model adjusted from all experimental data. (C) simplified model adjusted from reference condition; (D) simplified model adjusted from all experimental data. G: glucose, X: xylose and G2: cellobiose. Suffix *exp*: experimental and *pred*: predicted.

attributing physical meaning to the values of the model parameters was reported by Sin et al. [37]. Too large confidence interval was found during estimation due to strong correlation between some of the parameters. Although this does not compromise the model fit quality, it indicates that estimated parameters are not unique and should be considered as a convenient set of values that provides a good fit to the data [37].

4.2. Model simplification

Statistic parameter information as significance and confidence interval were not calculated for the original model using experimental data from reference condition because of ill-conditioning of Fisher Information Matrix. Modifying the model structure and/or increasing the information content of experimental data, are two ways to reduce the parametric correlation and improve the predictive ability of the model [37]. The proposed model simplification reduced the strong correlation between parameters (7 pairs of parameters remained strongly correlated after simplification) without affecting the model prediction capacity (Fig. 2C and rows OM and SM in Table 4); in fact, predictions of glucose concentration for sugar inhibition experiments were improved by the simplified model adjusted from a reference condition. Kinetic parameters estimated for simplified model (see Section 3.4) using reference condition and all experimental data are listed in columns 4 and 5 of Table 3.

4.3. Model verification

Although the original and simplified model has shown reasonable accuracy ($R^2 > 0.89$ and $R^2 > 0.93$, respectively) in describing the enzymatic hydrolysis data for reference condition (Fig. 2A and C), it has some difficulties in predicting cellobiose concentration under several evaluated experimental conditions and it has particular difficulty in describing the inhibition effect of sugar on enzyme (Table 4). This suggests that fitting the model only with a typical hydrolysis condition could be insufficient to achieve high fidelity in model predictions when operating in high-solids concentration.

The enzymatic hydrolysis for the reference condition was also described with accuracy by both original and simplified model adjusted from all experimental data (Fig. 2B and D). In addition, some sugar predictions were improved in respect to the adjusted model using a unique experimental condition. Comparisons of coefficient of determination (R^2) in Table 4 shows that xylose concentration predictions for all enzyme loading range (5–60 FPU/g-cellulose), cellobiose concentration predictions for enzyme loading >20 FPU/g-cellulose and glucose concentration prediction for different solids loading (10 and 20% w/v) and initial sugar concentrations (30 and 60 g/L-glucose, 10 g/L-cellobiose and 10 g/L-xylose) were improved by the simplified and original model adjusted from all experimental data.

Table 4Comparison of coefficients of determination (R^2) from simulation results for all experimental conditions.

Coefficient of determination (R^2)																						
Enzyme loading effect																						
Condition	5 FPU/g-C			10 FPU/g-C			15 FPU/g-C			20 FPU/g-C			25 FPU/g-C			30 FPU/g-C			60 FPU/g-C			
MODEL ^a	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	
OM	0.87	<0	0.83	0.98	0.89	0.96	0.97	0.44	0.79	0.97	0.12	0.82	0.97	<0	0.77	0.93	<0	0.71	0.89	<0	0.83	
OM-AE	0.93	0.50	0.99	0.97	0.67	0.76	0.98	0.22	0.95	0.98	0.94	0.95	0.98	0.95	0.91	0.90	0.72	0.91	0.88	0.84	0.88	
SM	0.95	<0	0.85	0.97	0.93	0.97	0.99	0.48	0.80	0.98	0.05	0.84	0.96	<0	0.80	0.82	<0	0.71	0.74	<0	0.83	
SM-AE	0.94	0.50	0.98	0.97	0.66	0.74	0.98	0.22	0.98	0.98	0.94	0.97	0.98	0.96	0.96	0.89	0.71	0.88	0.87	0.84	0.83	
Solid loading effect											Initial sugar concentration effect											
Condition	10% (w/v)			15% (w/v)			20% (w/v)			30G (g/L)			60G (g/L)			10G2 (g/L)			10X (g/L)			
MODEL ^a	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	G	G2	X	
OM	0.92	0.16	0.11	0.98	0.89	0.96	0.56	0.04	0.97	0.14	<0	<0	<0	<0	<0	0.83	<0	0.21	0.03	<0	0.58	
OM-AE	0.95	<0	0.46	0.97	0.67	0.76	0.62	0.61	0.79	0.86	0.09	0.05	0.84	<0	<0	0.96	<0	0.61	0.97	0.80	0.85	
SM	0.95	<0	0.18	0.97	0.93	0.97	0.56	0.13	0.97	0.80	<0	<0	0.76	<0	<0	0.17	<0	<0	0.92	0.49	0.75	
SM-AE	0.96	<0	0.58	0.97	0.66	0.74	0.62	0.61	0.84	0.85	<0	<0	0.83	<0	<0	0.95	0.83	0.73	0.97	0.80	0.86	

G = Glucose; G2 = Cellobiose; X = Xylose.

^a OM: Original model (with all 16 parameters); OM-AE: Original model adjusted with all experimental data; SM: Simplified model (with 10 parameters); SM-AE: Simplified model adjusted with all experimental data.

Model verification was conducted at various enzyme concentrations, solids loading and initial sugar concentration (i.e., glucose, cellobiose and xylose) and it is reported in the next sections for both original and simplified model.

4.3.1. Verification of enzyme concentration

The predicted glucose concentrations was over estimated by original and simplified model for the lowest (5 FPU/g-cellulose) and the highest (60 FPU/g-cellulose) enzyme loading evaluated (Fig. 3),

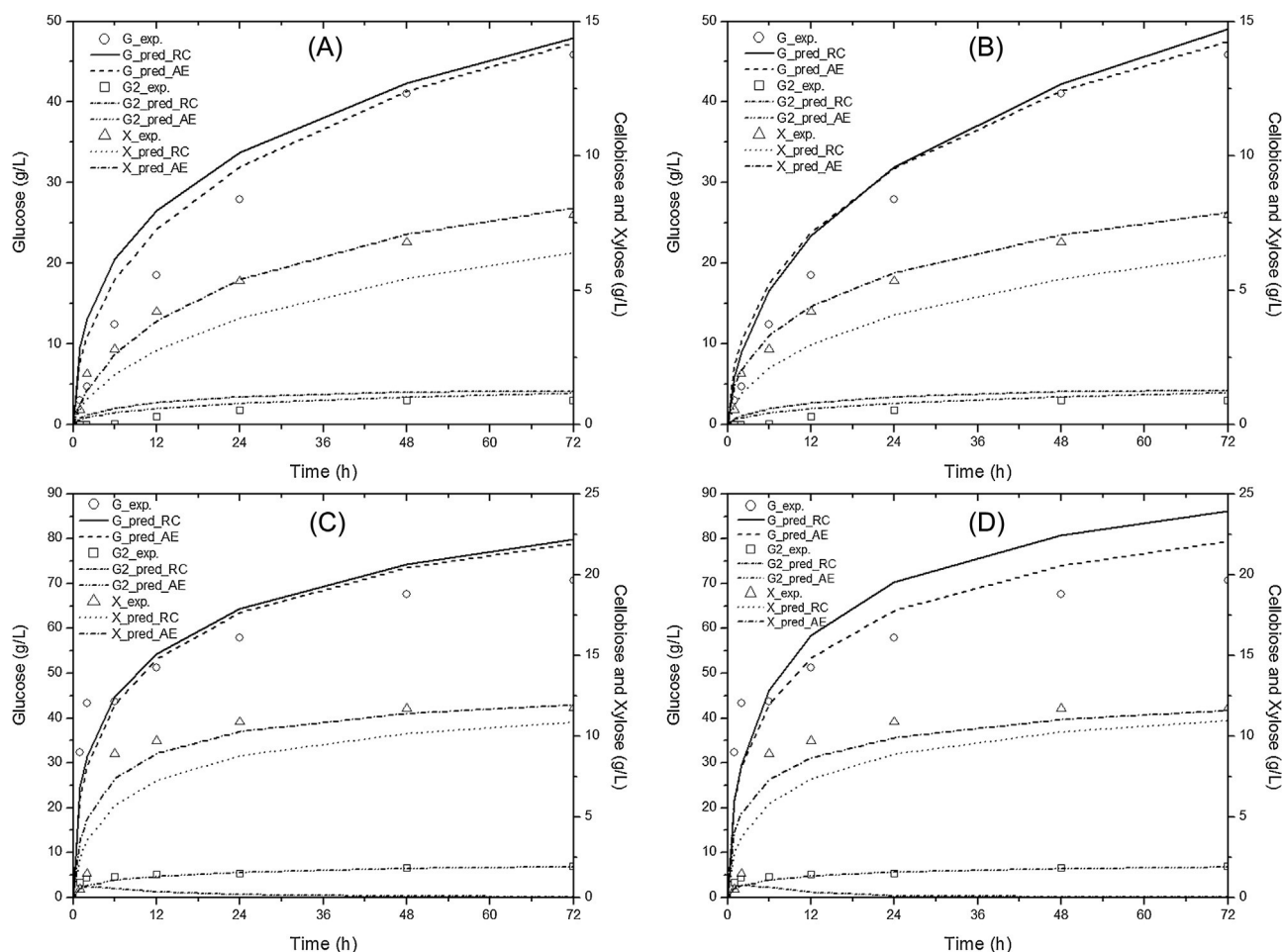


Fig. 3. Enzymatic hydrolysis of pretreated SCS using solid loading of 15% w/v and enzyme concentration of: (A) 5 FPU/g-cellulose for original model adjusted from reference condition (RC) and all experimental data (AE); (B) 5 FPU/g-cellulose for simplified model adjusted from RC and AE; (C) 60 FPU/g-cellulose for original model adjusted from RC and AE; (D) 60 FPU/g-cellulose for simplified model adjusted from RC and AE. Suffix *exp.*: experimental, *pred.*: predicted.

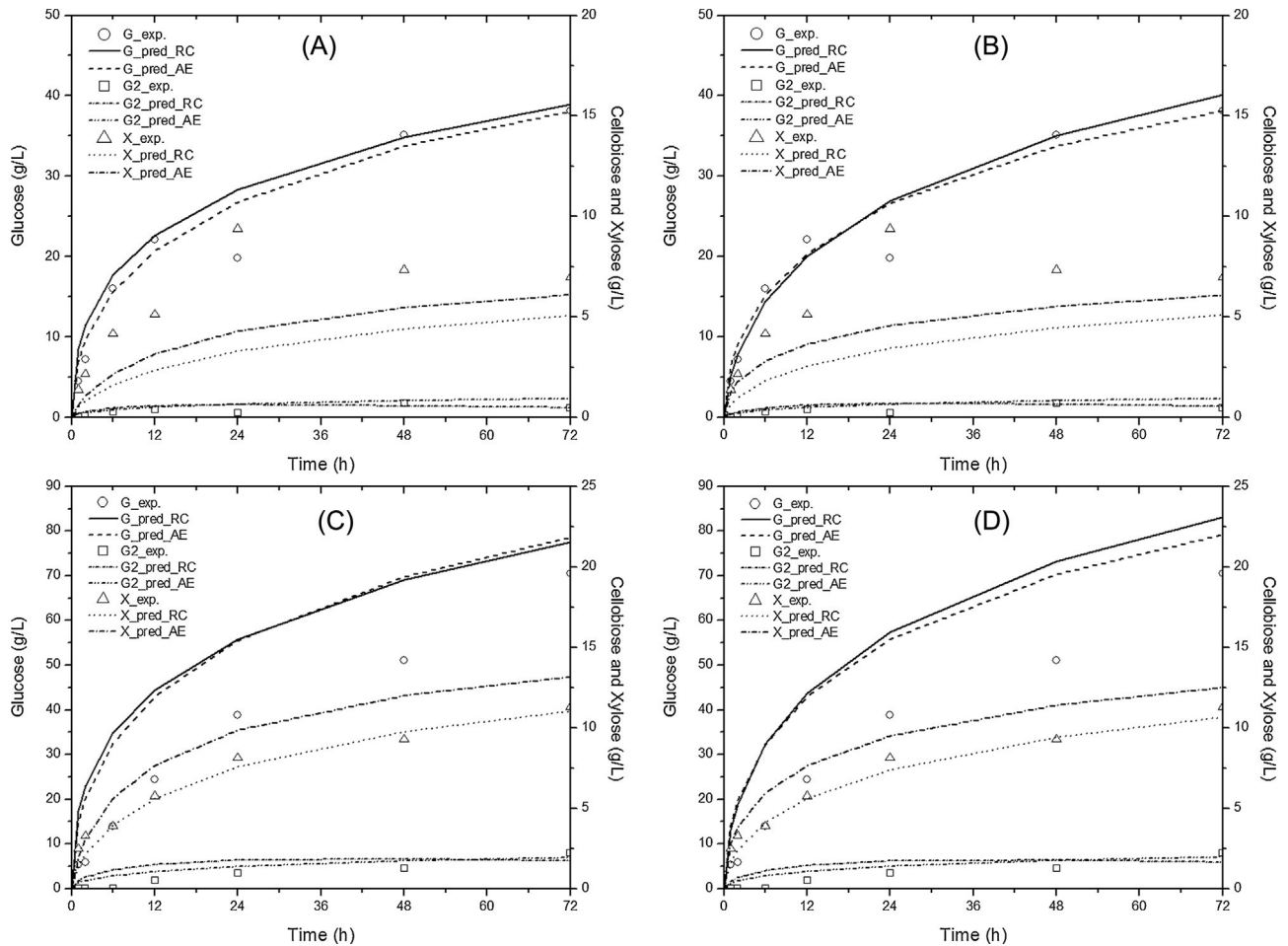


Fig. 4. Enzymatic hydrolysis of pretreated SCS using enzyme concentration of 10 FPU/g-cellulose and solids loading of: (A) 10% w/v of solids for original model adjusted from reference condition (RC) and all experimental data (AE); (B) 10% w/v of solids for simplified model adjusted from RC and AE; (C) 20% w/v of solids for original model adjusted from RC and AE; (D) 20% w/v of solids for simplified model adjusted from RC and AE.. Suffix *exp*: experimental, *pred*: predicted.

this might be explained by thermal and mechanical enzyme inactivation, unproductive adsorption enzyme, adsorption on lignin, and significant effect of end-product inhibition when working in high-solids concentration. The enzyme inactivation was shown to be more important for enzyme concentrations further away from reference condition (10 FPU/g-cellulose). Despite of the low yield (<2.1 g/L), cellobiose concentration was poorly predicted by the models, while xylose concentration was underestimated. Model predictability was much better from the models adjusted with all experimental data that improved the sugar predictions throughout the range of variation of the enzyme, especially cellobiose and xylose concentration for enzyme loading above 20 FPU/g-cellulose (rows OM-AE and SM-AE in Table 4 and Fig. 3).

4.3.2. Verification of solids loading

Experiments at high-solids concentration were conducted to investigate the effect of high substrate loading on the enzymatic hydrolysis of pretreated SCS. Glucose and xylose concentration were used to evaluate the model predictions. Cellobiose was not shown due to low concentration (<2.2 g/L) and the model limitations to describe this behavior (column of solid loading effect in Table 4). It is known that increasing solids level hinders the ability of the enzyme to reach the reaction site and this was not considered into the model, thus the original and simplified model overestimated glucose production for 20% of pretreated SCS (Fig. 4C and D). The models adjusted with all experimental conditions improved the xylose predictions for 10% (w/v) of solids loading, and the

glucose predictions for 20% (w/v) of solids loading. Still, glucose prediction for solid loading as high as 20% is not so good ($R^2 = 0.62$), this shows the difficulty of modeling the hydrolysis under high solids loading. Insoluble solids loading of 12–15% have been reported to be an upper limit for batch saccharification in stirred tank reactors [11].

4.3.3. Verification of sugar inhibition

Sugar inhibition of enzymes is a negative factor in the hydrolysis which becomes more important at high-solids concentrations due to increasing difficulty in diffusion of sugars away from catalytic sites [11]. Therefore the model was assessed to prove if it can describe correctly the effect of sugar inhibition. Enzyme inhibition for each sugar concentration is inversely proportional to its respective inhibition constant. Thus cellobiose inhibition in reaction r_1 should be stronger than glucose inhibition because cellobiose is the product of r_1 . Likewise glucose inhibition in reaction r_2 should be greater than cellobiose inhibition due to glucose is the direct product of cellulose saccharification. Comparing inhibition parameters from original model the results are consistent with this analysis, that is, $K_{11G2} < K_{11G}$ and $K_{21G} < K_{21G2}$ (columns 1 and 2 in Table 3). These conclusions cannot be extended to the simplified model because the sugar inhibition effects on r_1 , r_2 and r_4 have been lumped into a single set of inhibition parameters, and, as discussed in Section 4.1, one must be careful in assigning physical meaning to the values of model parameters.

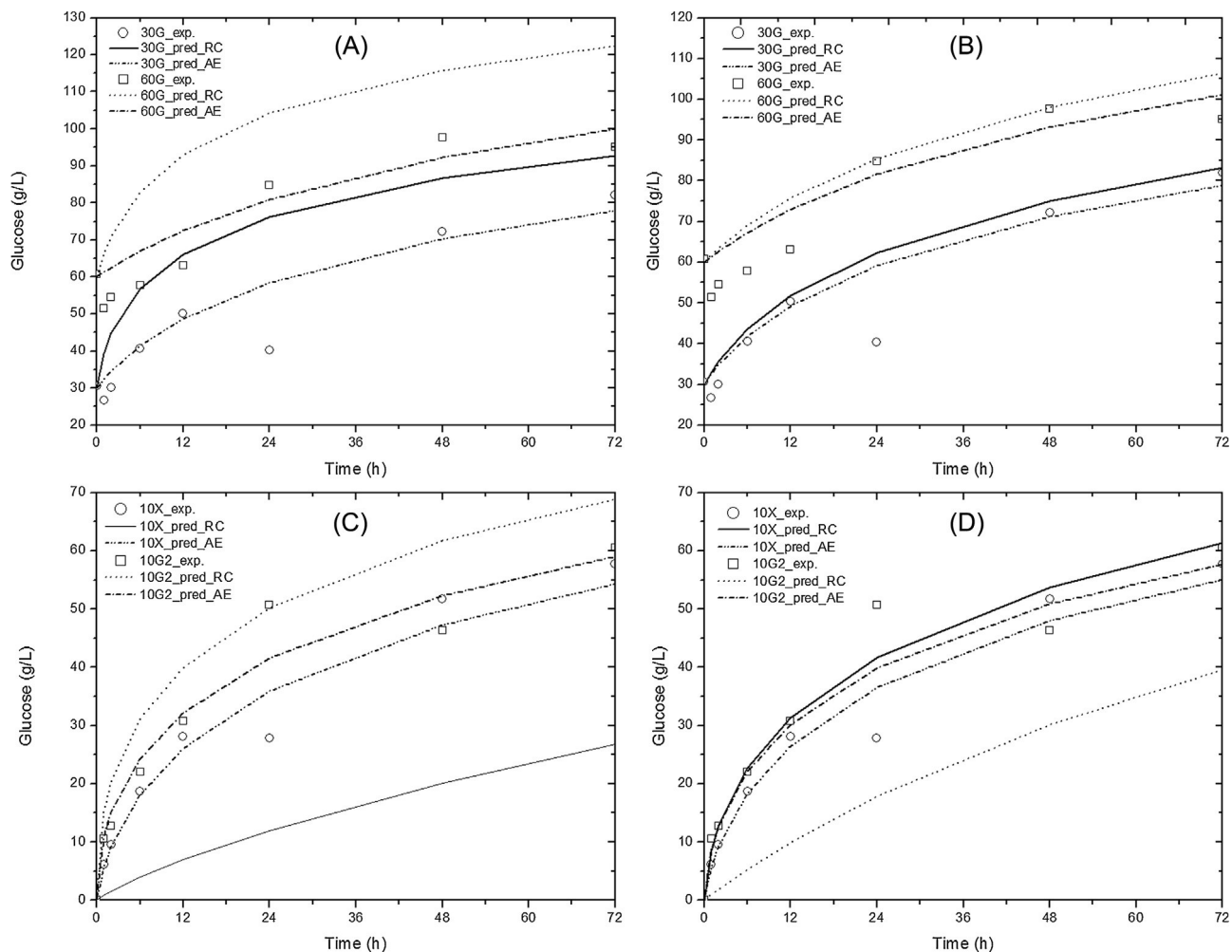


Fig. 5. Enzymatic hydrolysis (Glucose) of pretreated SCS using 15% (w/v, dry basis) solid loading, 10 FPU/g-cellulose enzyme concentration and initial concentration of: 30 g/L (30G) and 60 g/L (60G) of glucose from (A) original Model adjusted from reference condition (RC) and all experimental data (AE); (B) simplified model adjusted from RC and AE; 10 g/L of cellobiose (10G2) and 10 g/L of xylose (10X) from (C) original model adjusted from RC and AE; (D) simplified model adjusted from RC and AE.

In order to investigate the prediction capacity of the model under initial loading of sugars, experiments with initial concentration of 30 g/L (30G) and 60 g/L (60G) of glucose, 10 g/L (10G2) of cellobiose and 10 g/L (10X) of xylose were conducted. As it can be seen from the coefficient of determination, the original model (rows OM in column of initial sugar concentration effect of Table 4) did not represent the sugar inhibition effects; it is believed that mass transfer problems associated with operation at high-solids concentration were a critical factor in obtaining these results. Comparison between glucose experimental data in Figs. 2 and 5A showed that glucose is strongest inhibitor of glucose production. Also in Fig. 5 is shown that simplified and original model, adjusted with all experimental data, describes with better accuracy the sugar inhibition effect on glucose production ($R^2 > 0.83$). In fact, cellobiose and xylose predictions for cellobiose and xylose initial loading were improved with respect to the model (original or simplified) adjusted from a reference condition (see Table 4). This result suggests that using a unique reference condition to fit the model parameters, as is usually referred in the literature, is not sufficient to predict accurately the sugar inhibition effects in enzymatic hydrolysis at high-solids concentration. Although some limitations were maintained to describe cellobiose and xylose behavior under glucose inhibition, the original and simplified model showed acceptable performance in predicting the glucose concentration under the important inhibitory effects of glucose on enzymes in the complex conditions of high-solids concentration.

5. Conclusions

A semi-mechanistic multi-reaction kinetic model was proposed and verified for batch enzymatic hydrolysis at high-solids concentration of hydrothermally pretreated SCS. This model incorporated several factors, including: substrate reactivity (R_s), xylose formation and end-product inhibition. Parameter estimated from a reference high-solids condition showed a strong correlation. A proposed simplified model has shown to reduce the over-parameterization and improve the model fidelity. Original and simplified models presented a very close prediction capability when all available experimental data were used for parameter estimation. Although some limitations of the model to describe glucose concentration under solids loading as high as 20% (w/v) and the xylose and cellobiose concentration under inhibition effect of glucose (30 and 60 g/L), in general both original and simplified models could be used to describe with reasonable accuracy the enzymatic hydrolysis of SCS at high-solids concentration.

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